Plant Transfer Function of the Foam-Box Heater

500 mL cube, 2 cm EPS walls

Thermal Control System project

1 Goal and assumptions

The goal is to derive a first-order transfer function from heater duty cycle to temperature rise inside the box, starting from the energy balance.

- The box is a cube with an *inside* volume of $V = 500\,\text{mL} = 5 \times 10^{-4}\,\text{m}^3$.
- The wall thickness is $t = 2\,\text{cm}$ and the thermal conductivity is $\kappa = 0.033\,\text{W}\,\text{m}^{-1}\text{K}^{-1}$.
- The heater is a $R_{heater} = 6.2\,\Omega$ resistor driven by PWM from a supply of $V_s = 7.5\,\text{V}$.
- The air, the resistor and the inner foam are lumped into one body at a single temperature T.
- The box has no leaks (lid gaps, wire feedthroughs), and heat leaves only by conduction through the walls.

2 Energy balance

Let C_{th} be the thermal capacitance of the lumped body, R_{th} the thermal resistance to the room, P_{in} the heater power and T_{amb} the ambient temperature:

$$C_{th}\,\frac{dT}{dt} = P_{in}(t) - \frac{T(t) - T_{amb}}{R_{th}}. \tag{1}$$

Define the temperature rise $\theta(t) = T(t) - T_{amb}$. Because T_{amb} is constant, $d\theta/dt = dT/dt$, so

$$C_{th}\,\frac{d\theta}{dt} + \frac{\theta}{R_{th}} = P_{in}(t). \tag{2}$$

Taking the Laplace transform with zero initial conditions gives

$$\left(C_{th}s + \frac{1}{R_{th}}\right)\theta(s) = P_{in}(s) \quad \Longrightarrow \quad \frac{\theta(s)}{P_{in}(s)} = \frac{R_{th}}{R_{th}C_{th}\,s + 1}. \tag{3}$$

3 Heater input

The Arduino commands a duty cycle $D \in [0, 1]$. At 1 kHz or any PWM frequency far above the thermal bandwidth, only the average power matters:

$$P_{in} = D\,\frac{V_s^2}{R_{heater}}, \qquad P_{max} = \frac{7.5^2}{6.2} = 9.07\,\text{W}. \tag{4}$$

The plant from duty cycle to temperature rise is therefore

$$G(s) = \frac{\theta(s)}{D(s)} = \frac{(V_s^2/R_{heater})\,R_{th}}{R_{th}C_{th}\,s + 1} = \frac{K}{\tau s + 1}, \tag{5}$$

with

$$K = \frac{V_s^2}{R_{heater}}\,R_{th} \quad [^\circ\text{C per unit duty}], \qquad \tau = R_{th}C_{th} \quad [\text{s}]. \tag{6}$$

4 Geometry

The side length of the inside cube and its surface area are

$$a = \sqrt[3]{V} = \sqrt[3]{5 \times 10^{-4}} = 0.0794\,\mathrm{m}, \qquad A_{in} = 6a^2 = 0.0378\,\mathrm{m}^2. \tag{7}$$

With $t = 2\,\mathrm{cm}$, the outside cube has

$$a_{out} = a + 2t = 0.1194\,\mathrm{m}, \qquad A_{out} = 6a_{out}^2 = 0.0855\,\mathrm{m}^2. \tag{8}$$

5 Thermal resistance

Heat conducts through a hollow shell whose area grows from the inside surface to the outside surface. A simple representative area is the geometric mean:

$$A_{eff} = \sqrt{A_{in}\,A_{out}} = 0.0568\,\mathrm{m}^2. \tag{9}$$

The conduction resistance of the wall is

$$R_{th} = \frac{t}{\kappa\,A_{eff}} = \frac{0.02}{0.033 \times 0.0568} = 10.66\,^\circ\mathrm{C/W}. \tag{10}$$

For comparison, using A_{in} alone gives $16.0\,^\circ\mathrm{C/W}$ and using A_{out} alone gives $7.1\,^\circ\mathrm{C/W}$, so A_{eff} sits between the two bounds.

6 Thermal capacitance

The capacitance is the sum of the parts that follow the inside temperature.

Air. With $\rho_{air} = 1.2\,\mathrm{kg/m^3}$ and $c_{air} = 1005\,\mathrm{J/(kg\,K)}$:

$$C_{air} = \rho_{air}\,c_{air}\,V = 1.2 \times 1005 \times 5 \times 10^{-4} = 0.60\,\mathrm{J/^\circ C}. \tag{11}$$

Foam walls. The wall volume is $V_{wall} = a_{out}^3 - a^3 = 1.201 \times 10^{-3}\,\mathrm{m}^3$. For expanded polystyrene I assume $\rho_{EPS} = 25\,\mathrm{kg/m^3}$ and $c_{EPS} = 1300\,\mathrm{J/(kg\,K)}$:

$$m_{wall} = 25 \times 1.201 \times 10^{-3} = 30.0\,\mathrm{g}, \qquad C_{wall} = 0.0300 \times 1300 = 39.0\,\mathrm{J/^\circ C}. \tag{12}$$

The wall has a roughly linear temperature profile between the inside temperature and the room temperature, so on average it stores about half the heat it would store if it were entirely at the inside temperature. The effective wall capacitance is

$$C_{wall,eff} = \tfrac{1}{2}\,C_{wall} = 19.5\,\mathrm{J/^\circ C}. \tag{13}$$

Resistor. I assume a mass of $8\,\mathrm{g}$ and $c = 800\,\mathrm{J/(kg\,K)}$:

$$C_{res} = 0.008 \times 800 = 6.4\,\mathrm{J/^\circ C}. \tag{14}$$

Part	C (J/°C)
Air	0.60
Foam walls (effective)	19.5
Resistor	6.4
Total C_{th}	**26.5**

Table 1: Thermal capacitance budget.

7 Result

$$K = 9.07 \times 10.66 = 96.7\,^{\circ}\text{C per unit duty}, \qquad \tau = 10.66 \times 26.5 = 283\,\text{s}, \tag{15}$$

so

$$\boxed{G(s) = \frac{\theta(s)}{D(s)} \approx \frac{96.7}{283\,s + 1}} \qquad \text{or per watt:} \qquad \frac{\theta(s)}{P_{in}(s)} \approx \frac{10.66}{283\,s + 1}. \tag{16}$$

In absolute temperature the output is

$$T(t) = T_{amb} + \theta(t), \qquad T_{amb} \approx 25\,^{\circ}\text{C}. \tag{17}$$

The transfer function itself does not depend on the ambient temperature, because it describes the rise above ambient.

7.1 Duty cycle needed for a target temperature

At steady state $\theta_{ss} = K\,D_{ss}$, so

$$D_{ss} = \frac{T_{target} - T_{amb}}{K}. \tag{18}$$

For a 50 °C target, $D_{ss} = 25/96.7 \approx 26\%$. Full duty would aim for a rise of about 97 °C, which is far above what EPS foam tolerates, so the duty cycle should be limited in both the simulation and the Arduino code.

7.2 Effect of the outside air film

The conduction-only model neglects the convective film on the outside of the box. With a typical combined coefficient $h \approx 8\,\text{W/(m}^2\text{K)}$:

$$R_{film} = \frac{1}{h\,A_{out}} = \frac{1}{8 \times 0.0855} = 1.46\,^{\circ}\text{C/W}, \tag{19}$$

which gives $R_{th} = 12.12\,^{\circ}\text{C/W}$, $K \approx 110$ and $\tau \approx 321\,\text{s}$. The result is therefore best quoted as $K \approx 97$–110 and $\tau \approx 280$–$320\,\text{s}$.

8 Limitations

- The model is a single node. In reality the resistor body runs hotter than the air.
- Leaks reduce R_{th} and therefore K and τ.
- The foam density, specific heat and the resistor mass are estimates.
- The DS18B20 sensor lag is not included here; it adds a second first-order factor $1/(\tau_s s + 1)$ in the feedback path.
- A measured open-loop step response (fixed duty, logged temperature) should be used to confirm K and τ: K is the final rise divided by the duty, and τ is the time to reach 63% of the final rise.